

Data Designer + Switchyard

Measured routing experiments

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Result. Task-hinted routing achieved 100% exact-answer accuracy at an estimated cost of \$0.021838, compared with 100% at \$0.034776 for strong-only. This is a 37.2% cost reduction. Weak-only achieved 85% at \$0.000219 for weak-only.

Experiment design

The benchmark keeps Data Designer’s schema, seeding, structured-output handling, and async execution unchanged. A single OpenAI-compatible provider points to five logical Switchyard routes:

- **dd-weak:** direct weak model
- **dd-strong:** direct strong model
- **dd-hinted:** Data Designer supplies an easy/hard hint and Switchyard selects the matching tier
- **dd-smart:** optional LLM-classifier route used during calibration
- **dd-mixed:** seeded 35% strong / 65% weak routing for creative generation

The quality set contains 20 exact-answer tasks: sixteen simple transformations or arithmetic tasks and four combinatorics or number-theory tasks. The diversity run generates 24 responses to one constrained support-ticket prompt.

Models

Role	Model
Weak	nvidia/nvidia/Nemotron-3-Nano-30B-A3B
Strong	aws/anthropic/bedrock-claude-sonnet-4-6
Classifier	nvidia/deepseek-ai/deepseek-v4-flash

Quality and estimated cost

Route	All	Easy	Hard	S / W	Classifier	Est. cost
Weak	85%	100%	25%	- / -	0	\$0.000219
Strong	100%	100%	100%	- / -	0	\$0.034776
Hinted	100%	100%	100%	4 / 16	0	\$0.021838

Cost per correct row was \$0.000013 for weak-only, \$0.001739 for strong-only, and \$0.001092 for task-hinted routing.

Diversity

Route	Vocabulary	Distinct-1	Distinct-2	Mean Jaccard	Duplicates
Weak	212	0.328	0.754	0.143	0%
Mixed	223	0.308	0.614	0.198	0%

Higher distinct-n scores and vocabulary are better. Lower pairwise Jaccard similarity and duplicate rate indicate more varied outputs. The mixed route increased vocabulary, but worsened both distinct-n and pairwise similarity. This small run does not support model mixing as a diversity improvement.

Interpretation

The experiment tests Switchyard as an operational layer below Data Designer, not as a replacement for Data Designer model aliases. The useful addition is per-request backend selection, one logical endpoint, centralized usage accounting, and the ability to change the model mixture without changing dataset configuration. The prompt marker used here should become explicit request metadata in a production integration.

The benchmark is intentionally small. Before production use, repeat it with domain-specific validators, more samples, stable endpoint conditions, and a route policy trained or calibrated on representative prompts.